



Operating Instructions

**AIM-9874, AIM-9875, AIM-9876,
AIM-9877, AIM-9878, & AIM-9879
HUMIDITY / ENVIRONMENT
CHAMBER**



Aimil Ltd.

Instrumentation & Technologies



Aimil Ltd.

Instrumentation & Technologies

Introduction

Thank you for purchasing this Humidity / Environmental Chamber from Aimil. Humidity / Environmental Chambers facilitate the scientists to process the samples at a controlled environmental condition. When a sample undergoes ageing in a controlled environment, its physical properties are predictive in nature and thus the chambers are very useful for characterizing the materials.

Scope

The scope of this manual covers the Operating Instructions for various models of Humidity / Environmental chambers as per specification details contained in this manual.



Caution

The Humidity / Environmental chamber is suitable for indoor operation only and is used strictly in accordance with the details contained in this manual. The Humidity / Environmental Chamber is required to be placed on a level ground in the laboratory.

The Humidity / Environmental chambers supplied by Aimil are double walled in construction, with the complete inner chamber made of stainless steel. The outer body is made of a mild steel sheet which is powder coated. The space between the inner and outer wall is filled with high quality glass wool to prevent the loss of chamber temperature.

Key Feature

- Ø Double Walled Construction
- Ø High Quality Insulation
- Ø Precise Control and maintenance of Temperature and Humidity

Specifications :

Instrument and Manual Identification

The instrument is identified by a catalogue number and serial number on the serial number plate on the rear panel of the instrument. This manual applies to the instrument with catalogue indicated on the title page.

Product Specifications

Catalogue No.	Chamber Size Inches (mm)		
	W	H	D
AIM - 9874	18" (455)	18" (455)	28" (710)
AIM - 9875	24" (610)	24" (610)	24" (610)
AIM - 9876	24" (610)	24" (610)	36" (910)
AIM - 9877	18" (455)	18" (455)	28" (710)
AIM - 9878	24" (610)	24" (610)	24" (610)
AIM - 9879	24" (610)	24" (610)	36" (910)

**Aimil Ltd.**

Instrumentation & Technologies

Operating Specifications

Parameters	Models 9874,75 & 76	Models 9877,78 & and 79
Nominal Mains operating Voltage	230 V +/- 10 % 1*	
Mains operating Frequency	50 Hz	
Maximum Electrical Load	1.5 kW	2.5 kW
Control Type	Digital PID	
Temperature Range	Ambient +5° to 60° C +/- 1°C	10°C to 45°C +/- 1°C
Humidity Range (RH)	50 to 95% +/- 3%	50 to 95% +/- 4%
Continuous working temperature	Ambient to 50° C	10 to 45° C
Continuous Humidity (RH)	50 to 92 +/- 5%	

Ordering Information :

Sl.No.	Cat. No.	Product Description
1	AIM-9874	Humidity & Temperature control cabinet (only heating) , Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 455 X 455 X 710 mm
2	AIM-9875	Humidity & Temperature control cabinet (only heating) , Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 610 X 610 X 610 mm
3	AIM-9876	Humidity & Temperature control cabinet (only heating) , Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 610 X 610 X 910 mm
4	AIM-9877	Environmental Test Chamber "Eco Friendly" with heating and cooling , Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 455 X 455 X 710 mm
5	AIM -9878	Environmental Test Chamber "Eco Friendly" with heating and cooling, Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 610 X 610 X 610 mm
6	AIM-9879	Environmental Test Chamber "Eco Friendly" with heating and cooling, Temperature range Ambient +5° to 60° C +/- 1°C Relative Humidity Ambient to 95% +/- 3%, Double walled, Inner chamber made of SS 304, Digital Controller cum indicator, Size 610 X 610 X 910 mm



Aimil Ltd.

Instrumentation & Technologies

Unpacking

Carefully inspect the package containing the Humidity / Environmental chamber for any external damage during transit. If there are signs of external damage, inform your AIMIL salesperson or AIMIL authorised distributor through whom you purchased the equipment. Also, please mention the damage on the transporter's receipt.

If everything is in order, after informing your AIMIL service engineer or authorised AIMIL distributor, carefully unpack the container and relieve all the packing materials.

If you observe any damage / dents, inform Aimil or Distributors through whom the humidity chamber was purchased giving details of your purchase order and Aimil's invoice / packing list details.



Note :

If the humidity / environmental chamber is required to be returned to Aimil for future services, the instrument should be packed in a similar manner to the manner it was received in. Hence, it may be convenient to retain the original packing for future use.

Installation :

The Humidity / Environmental chamber is intended for indoor usage only. Do not install the machine in a place open to the sky or where direct sunlight or water may be able to fall on it. The Humidity / Environmental chamber should be installed on a firm, level ground. Leave at least 6" space around the Humidity / Environmental chamber. The power should be connected to the Humidity Chamber through a 15A power socket, wired through a fuse.

For the fuse rating, please check the details appropriate to the model you have purchased. Ensure that the location where you install the humidity / environmental chamber is kept having ample space for the user to work with. The space in front should be more than the swing of the chamber door to open fully, such that the user can conveniently and comfortably place the samples for processing.

Installation Requirements

Location :

The Humidity / Environmental chamber is intended for indoor operation only. The location where the Humidity / Environmental chamber is to be installed should be well ventilated with an ambient temperature range between 5°C and 35°C and a relative humidity of 85% or less. The Location should be free from Dust, Corrosive Gases, Vibration and Shocks. It should also be sheltered from direct sunlight and rain and placed safely away from intense Acoustic noise and from Electromagnetic & Radio Frequency Interferences.

Placement of Instrument :

Care should be taken when lifting or moving the machine. A clearance of at least 150mm on the sides and back is needed for proper ventilation and operational convenience and user comfort. The louvers in the underside of the machine and the grill in the rear must be kept clear of obstructions.



Aimil Ltd.

Instrumentation & Technologies



Note :

The clearance is essential to ensure proper ventilation for the heat generating components inside the instrument.

The instrument is provided with soft, insulating and non scratching feet and a three conductor integral power cord with a fitted 3 pin 15A plug — which when plugged into an appropriate power outlet grounds the instrument. (Panel, cabinet and chassis)

POWER REQUIREMENT

The Humidity / Environmental chamber requires a single phase 50 Hz power supply of 230V, in a 3 contact 15A AC power outlet with its ground contact properly earthed, conforming to the relevant specifications of the Bureau of Indian Standards (BIS).

The maximum power consumption of the instrument is stated in the product specification page. Please refer to the page for details and ensure proper wiring.

CONNECTING POWER TO THE EQUIPMENT :

Connect the supply to the Humidity / Environmental chamber through the wall outlet as specified above. For routine operation, please refer to the operation of the Humidity / Environmental chamber.

The Control Panel :

The control panel consists of :

Sl.No.	For Models 9874, 9875 and 9876	For Models 9877, 9878 and 9879
1	Digital temperature controller for Dry temperature & Wet Temperature.	Digital temperature controller for Dry temperature & Wet Temperature.
2	Mains Switch	Mains Switch.
3	Heating Switch	Heating Switch
4	Humidity Switch	Humidity Switch.
5	Water Level Indicator	Water Level Indicator.
6	Water Tank, float valve, Water outlet.	Water Tank, Float Valve, Water outlet.
7	Water bath, SS with blower and spray pipe fitted with 1.25kW / 1.5 kW heater for steam generation mounted on a sheet metal frame.	Water bath, SS with blower and spray pipe fitted with 1.8kW / 2.5 kW heater for steam generation mounted on a sheet metal frame.
8		Cooling Switch.
9		Through Stabilizer.
10		Digital Volt meter.
11		Input & Output Switch.

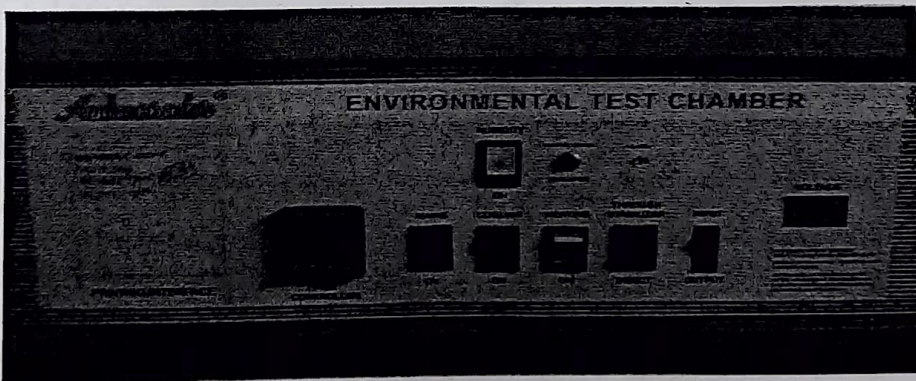


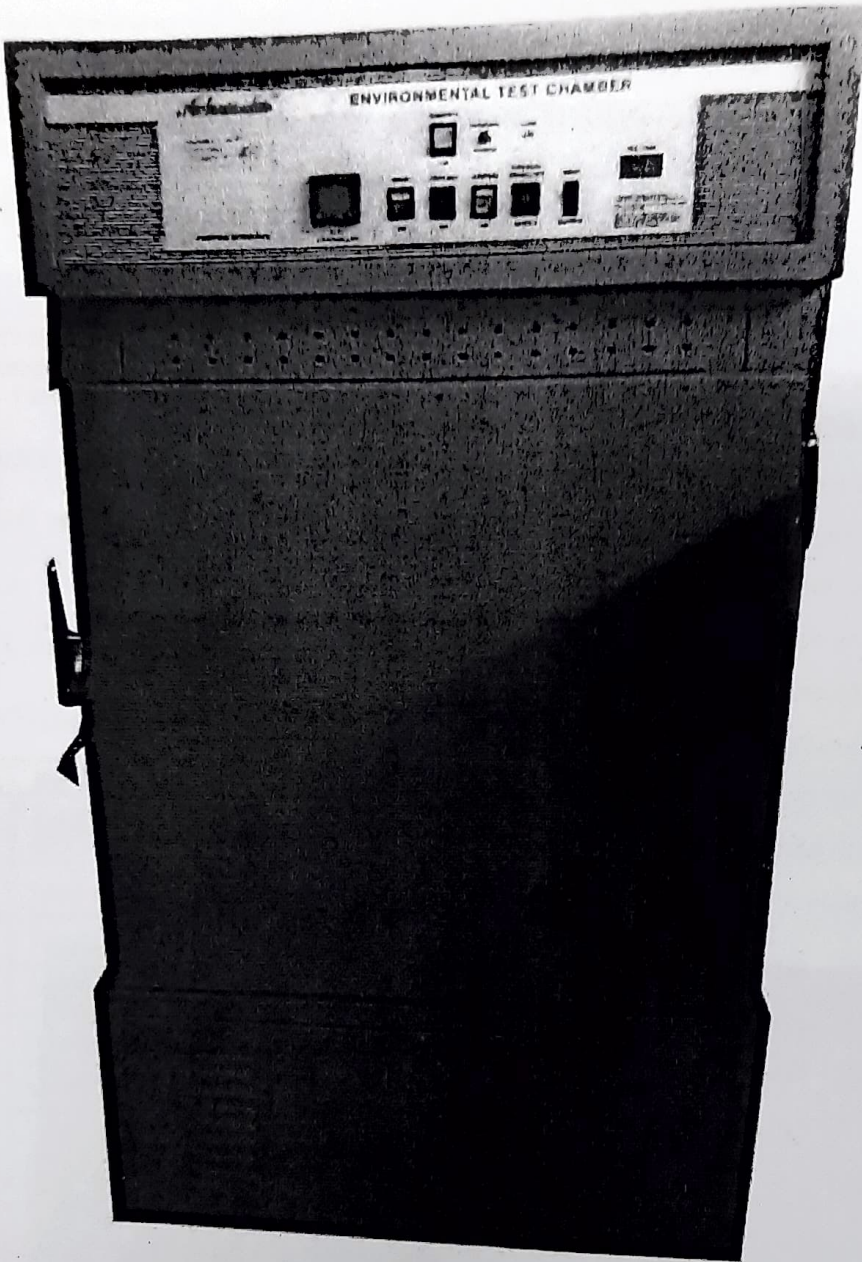
Operation of Humidity / Environment Chamber

NOTE:

There are no major differences in the operation of the models provided with or without cooling systems. Thus, the operation for all the models is very similar in nature, excepting that the models with cooling system incorporated can be used for temperatures at or below ambient. Once the required temperature is set using the temperature controller, the control is automatic.

- 1) Before attempting to operate the chamber, fill the water inside the chamber.
- 2) Power the chamber up using the Mains ON switch, the pilot indicator will glow, indicating that the power to the chamber is active.
- 3) On the PID controller, while pressing the page button, press the up/down scroll buttons as appropriate to set the required temperature. To set the humidity level, press the page button and repeat the process. Then press the page button to return to the home page.
- 4) After the desired temperature & humidity values switch the cooling, heating & humidity on.
- 5) The humidity can be checked / measured by the wet and dry thermometer have been set, provided along with the instrument.
- 6) The indicators will blink, indicating that the set point is achieved, as soon as the required Humidity is reached and maintained.







Aimil Ltd.

Instrumentation & Technologies

